$$\frac{\mathrm{d}f_3}{\mathrm{d}t} = \frac{1}{L} u_{source} - \frac{R_{el}}{L} f_3 - \frac{K}{L} f_7$$

$$\frac{\mathrm{d}f_7}{\mathrm{d}t} = \frac{K}{J} f_3 - \frac{R_{bearing}}{J} f_7 + \frac{D}{2} m \frac{\mathrm{d}f_{11}}{\mathrm{d}t} + \frac{D}{2} mg$$

$$f_{11} = -\frac{D}{2} f_7$$